QUE :- To study, implement, and analyze various Disk Scheduling Algorithms (**FCFS, SSTF, SCAN, and C-SCAN**) and evaluate their performance based on the **Total Head Movement** required to satisfy a given set of I/O requests.

CODE :

index.html

Disk Scheduling Algorithms > index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Disk Scheduling Algorithms Lab</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>
        @import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;600;700&display=swap');
        body { font-family: 'Inter', sans-serif; }
        .algorithm-card { transition: all 0.3s ease; }
        .algorithm-card:hover { transform: translateY(-5px); }
        .chart-container { height: 250px; width: 100%; position: relative; }
    </style>
</head>
<body class="bg-gray-50 text-gray-900">

    <!-- Navigation/Header -->
    <nav class="bg-white shadow-sm border-b sticky top-0 z-10">
        <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
            <div class="flex justify-between h-16 items-center">
                <div class="flex items-center space-x-2">
                    <div class="bg-blue-600 p-2 rounded-lg">
                        <svg class="w-6 h-6 text-white" fill="none" stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round" stroke-linejoin="round" stroke-width="2
                    </div>
                    <span class="text-xl font-bold tracking-tight">DiskSched<span class="text-blue-600">Pro</span></span>
                </div>
                <div class="hidden md:flex space-x-8">
                    <a href="#learn" class="hover:text-blue-600 font-medium">Learn</a>
                    <a href="#simulator" class="hover:text-blue-600 font-medium">Simulator</a>
                </div>
            </div>
        </div>
    </nav>

    <main class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

        <!-- Hero Section -->
        <section id="learn" class="mb-12">
            <div class="text-center mb-10">
                <h1 class="text-4xl font-extrabold text-gray-900 mb-4">Disk Scheduling Algorithms</h1>
                <p class="text-lg text-gray-600 max-w-2xl mx-auto">Visualize and calculate the <strong>Total Head Movement</strong> required to satisfy disk I/O requests.</p>
            </div>

            <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-4">
                <div class="algorithm-card bg-white p-5 rounded-xl shadow-sm border border-gray-100">
                    <h3 class="font-bold text-blue-600">FCFS</h3>
                    <p class="text-xs text-gray-500 mb-2 uppercase tracking-tighter">First Come First Serve</p>
                    <p class="text-gray-600 text-xs">Services requests in strict order of arrival. Simple but high head movement.</p>
                </div>
                <div class="algorithm-card bg-white p-5 rounded-xl shadow-sm border border-gray-100">
                    <h3 class="font-bold text-green-600">SSTF</h3>
                    <p class="text-xs text-gray-500 mb-2 uppercase tracking-tighter">Shortest Seek Time First</p>
                    <p class="text-gray-600 text-xs">Greedy approach. Picks the closest request to the current head position.</p>
                </div>
                <div class="algorithm-card bg-white p-5 rounded-xl shadow-sm border border-gray-100">
                    <h3 class="font-bold text-orange-600">SCAN</h3>
                    <p class="text-xs text-gray-500 mb-2 uppercase tracking-tighter">Elevator Algorithm</p>
                    <p class="text-gray-600 text-xs">Moves head from one end to the other, servicing all requests in between.</p>
                </div>
                <div class="algorithm-card bg-white p-5 rounded-xl shadow-sm border border-gray-100">
                    <h3 class="font-bold text-red-600">C-SCAN</h3>
                    <p class="text-xs text-gray-500 mb-2 uppercase tracking-tighter">Circular SCAN</p>
                    <p class="text-gray-600 text-xs">Provides uniform waiting time by only servicing requests in one direction.</p>
                </div>
            </div>
        </section>

```

index.html

Disk Scheduling Algorithms > index.html > ...

```html
<html lang="en">
<body class="bg-gray-50 text-gray-900">
    <main class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

        <!-- Simulator Configuration -->
        <section id="simulator" class="bg-white rounded-2xl shadow-lg border border-gray-200 mb-8">
            <div class="p-6 border-b border-gray-100 flex flex-col md:flex-row md:items-end gap-4">
                <div class="flex-grow">
                    <label class="block text-xs font-bold text-gray-500 uppercase mb-1">Request Queue (e.g., 98, 183, 37)</label>
                    <input id="queue-input" type="text" value="98, 183, 37, 122, 14, 124, 65, 67"
                           class="w-full px-3 py-2 rounded-lg border border-gray-300 focus:ring-2 focus:ring-blue-500 outline-none">
                </div>
                <div class="w-24">
                    <label class="block text-xs font-bold text-gray-500 uppercase mb-1">Start Head</label>
                    <input id="head-input" type="number" value="53" class="w-full px-3 py-2 rounded-lg border border-gray-300">
                </div>
                <div class="w-24">
                    <label class="block text-xs font-bold text-gray-500 uppercase mb-1">Max Cyl.</label>
                    <input id="max-input" type="number" value="200" class="w-full px-3 py-2 rounded-lg border border-gray-300">
                </div>
                <div class="w-32">
                    <label class="block text-xs font-bold text-gray-500 uppercase mb-1">Direction</label>
                    <select id="direction-input" class="w-full px-3 py-2 rounded-lg border border-gray-300">
                        <option value="right">Right (High)</option>
                        <option value="left">Left (Low)</option>
                    </select>
                </div>
                <button onclick="simulate()" class="bg-blue-600 hover:bg-blue-700 text-white px-8 py-2 rounded-lg font-bold transition-all shadow-md active:scale-95">
                    Run Simulation
                </button>
            </div>
        </section>

        <!-- Results Grid -->
        <div id="visual-grid" class="grid grid-cols-1 md:grid-cols-2 gap-8">
            <!-- FCFS Chart -->
            <div class="bg-white p-6 rounded-2xl border border-gray-200 shadow-sm opacity-50 transition-opacity duration-500" id="card-FCFS">
                <div class="flex justify-between items-center mb-4">
                    <h3 class="font-bold text-blue-600">FCFS</h3>
                    <div class="text-right">
                        <p class="text-[10px] text-gray-400 uppercase font-bold">Total Head Movement</p>
                        <span class="text-sm font-mono font-bold text-blue-800" id="move-FCFS">0</span>
                    </div>
                </div>
                <div class="chart-container"><canvas id="chart-FCFS"></canvas></div>
            </div>

            <!-- SSTF Chart -->
            <div class="bg-white p-6 rounded-2xl border border-gray-200 shadow-sm opacity-50 transition-opacity duration-500" id="card-SSTF">
                <div class="flex justify-between items-center mb-4">
                    <h3 class="font-bold text-green-600">SSTF</h3>
                    <div class="text-right">
                        <p class="text-[10px] text-gray-400 uppercase font-bold">Total Head Movement</p>
                        <span class="text-sm font-mono font-bold text-green-800" id="move-SSTF">0</span>
                    </div>
                </div>
                <div class="chart-container"><canvas id="chart-SSTF"></canvas></div>
            </div>

            <!-- SCAN Chart -->
            <div class="bg-white p-6 rounded-2xl border border-gray-200 shadow-sm opacity-50 transition-opacity duration-500" id="card-SCAN">
                <div class="flex justify-between items-center mb-4">
                    <h3 class="font-bold text-orange-600">SCAN</h3>
                    <div class="text-right">
                        <p class="text-[10px] text-gray-400 uppercase font-bold">Total Head Movement</p>
                        <span class="text-sm font-mono font-bold text-orange-800" id="move-SCAN">0</span>
                    </div>
                </div>
                <div class="chart-container"><canvas id="chart-SCAN"></canvas></div>
            </div>
```

index.html

Disk Scheduling Algorithms > index.html > ...

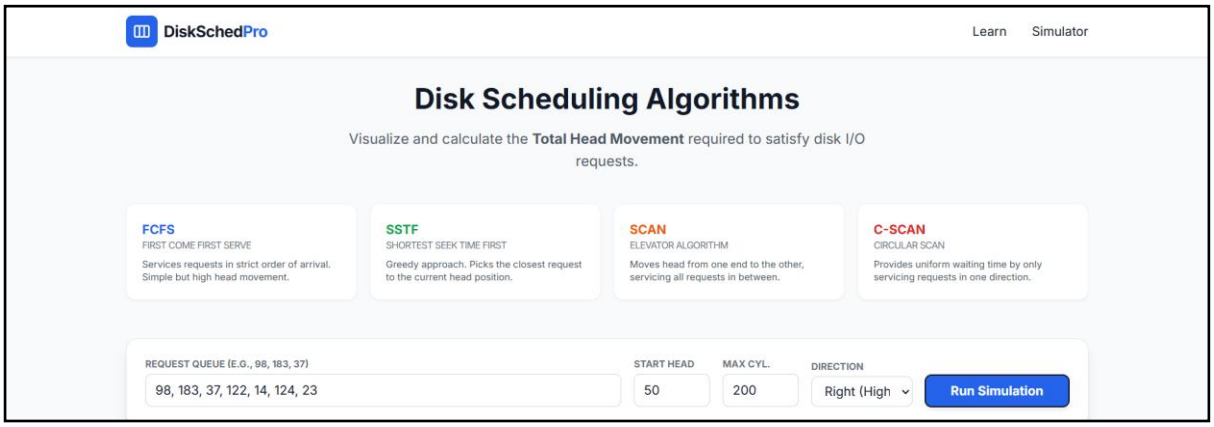
```html
<html lang="en">
<body class="bg-gray-50 text-gray-900">
    <main class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">
        <div id="visual-grid" class="grid grid-cols-1 md:grid-cols-2 gap-8">

            <!-- C-SCAN Chart -->
            <div class="bg-white p-6 rounded-2xl border border-gray-200 shadow-sm opacity-50 transition-opacity duration-500" id="card-CSCAN">
                <div class="flex justify-between items-center mb-4">
                    <h3 class="font-bold text-red-600">C-SCAN</h3>
                    <div class="text-right">
                        <p class="text-[10px] text-gray-400 uppercase font-bold">Total Head Movement</p>
                        <span class="text-sm font-mono font-bold text-red-800" id="move-CSCAN">0</span>
                    </div>
                </div>
                <div class="chart-container"><canvas id="chart-CSCAN"></canvas></div>
            </div>
        </div>
    </main>

    <script>
        let charts = {};

        function simulate() {
            const rawQueue = document.getElementById('queue-input').value;
            const head = parseInt(document.getElementById('head-input').value);
            const max = parseInt(document.getElementById('max-input').value);
            const dir = document.getElementById('direction-input').value;

            if (!rawQueue) return;
            const queue = rawQueue.split(',').map(n => parseInt(n.trim())).filter(n => !isNaN(n));

            const results = [
                runFCFS([...queue], head),
                runSSTF([...queue], head),
                runSCAN([...queue], head, max, dir),
                runCSCAN([...queue], head, max, dir)
            ];

            results.forEach(res => {
                const id = res.name.replace('-', '');
                document.getElementById(`card-${id}`).classList.remove('opacity-50');
                document.getElementById(`move-${id}`).innerText = `${res.movement} Cylinders`;
                renderIndividualChart(id, res.sequence, res.color, max);
            });
        }

        function renderIndividualChart(id, sequence, color, max) {
            const ctx = document.getElementById(`chart-${id}`).getContext('2d');

            if (charts[id]) charts[id].destroy();

            charts[id] = new Chart(ctx, {
                type: 'line',
                data: {
                    labels: sequence.map((_, i) => i),
                    datasets: [{
                        label: 'Head Position',
                        data: sequence,
                        borderColor: color,
                        backgroundColor: color + '22',
                        fill: false,
                        tension: 0,
                        pointRadius: 5,
                        pointHoverRadius: 8,
                        borderWidth: 3
                    }]
                },
                options: {
                    indexAxis: 'y',
                    responsive: true,
                    maintainAspectRatio: false,
                    scales: {
                        x: {
                            min: 0,
                            max: max,
                            title: { display: true, text: 'Cylinder' }
                        },
                        y: {
                            reverse: true,
                            title: { display: true, text: 'Order of Service' },
                            ticks: { display: false }
                        }
                    },
                    plugins: {
                        legend: { display: false },
                        tooltip: {
```

index.html

Disk Scheduling Algorithms > index.html > ...

```html
<html lang="en">
<body class="bg-gray-50 text-gray-900">
    <script>
        function renderIndividualChart(id, sequence, color, max) {
                options: {
                    scales: {
                        x: {
                            min: 0,
                            max: max,
                            title: { display: true, text: 'Cylinder' }
                        },
                        y: {
                            reverse: true,
                            title: { display: true, text: 'Order of Service' },
                            ticks: { display: false }
                        }
                    },
                    plugins: {
                        legend: { display: false },
                        tooltip: {
                            callbacks: {
                                label: (context) => `Cylinder: ${context.parsed.x}`
                            }
                        }
                    }
                }
            });
        }

        function runFCFS(q, head) {
            let sequence = [head, ...q];
            let move = 0;
            for(let i=0; i<sequence.length-1; i++) move += Math.abs(sequence[i] - sequence[i+1]);
            return { name: "FCFS", sequence, movement: move, color: '#3b82f6' };
        }

        function runSSTF(q, head) {
            let sequence = [head];
            let current = head;
            let tempQ = [...q];
            while(tempQ.length > 0) {
                let closest = tempQ.reduce((prev, curr) => Math.abs(curr - current) < Math.abs(prev - current) ? curr : prev);
                sequence.push(closest);
                tempQ = tempQ.filter(item => item !== closest);
                current = closest;
            }
            let move = 0;
            for(let i=0; i<sequence.length-1; i++) move += Math.abs(sequence[i] - sequence[i+1]);
            return { name: "SSTF", sequence, movement: move, color: '#10b981' };
        }

        function runSCAN(q, head, max, dir) {
            let left = q.filter(n => n < head).sort((a,b) => b-a);
            let right = q.filter(n => n > head).sort((a,b) => a-b);
            let sequence = [head];
            if(dir === 'right') sequence = [head, ...right, max - 1, ...left];
            else sequence = [head, ...left, 0, ...right];
            let move = 0;
            for(let i=0; i<sequence.length-1; i++) move += Math.abs(sequence[i] - sequence[i+1]);
            return { name: "SCAN", sequence, movement: move, color: '#f59e0b' };
        }

        function runCSCAN(q, head, max, dir) {
            let left = q.filter(n => n < head).sort((a,b) => a-b);
            let right = q.filter(n => n > head).sort((a,b) => a-b);
            let sequence = [head];
            if(dir === 'right') sequence = [head, ...right, max - 1, 0, ...left];
            else sequence = [head, ...left, 0, max - 1, ...right];
            let move = 0;
            for(let i=0; i<sequence.length-1; i++) move += Math.abs(sequence[i] - sequence[i+1]);
            return { name: "C-SCAN", sequence, movement: move, color: '#ef4444' };
        }
    </script>
</body>
</html>
```

OUTPUT :


DiskSchedPro
Learn
Simulator
Disk Scheduling Algorithms
Visualize and calculate the Total Head Movement required to satisfy disk I/O requests.
FCFS
FIRST COME FIRST SERVE
Services requests in strict order of arrival. Simple but high head movement.
SSTF
SHORTEST SEEK TIME FIRST
Greedy approach. Picks the closest request to the current head position.
SCAN
ELEVATOR ALGORITHM
Moves head from one end to the other, servicing all requests in between.
C-SCAN
CIRCULAR SCAN
Provides uniform waiting time by only servicing requests in one direction.
REQUEST QUEUE (E.G., 98, 183, 37)
98, 183, 37, 122, 14, 124, 23
START HEAD
50
MAX CYL.
200
DIRECTION
Right (High
Run Simulation

DiskSchedPro
Learn
Simulator
FCFS
TOTAL HEAD MOVEMENT
683 Cylinders
SSTF
TOTAL HEAD MOVEMENT
205 Cylinders
SCAN
TOTAL HEAD MOVEMENT
334 Cylinders
C-SCAN
TOTAL HEAD MOVEMENT
385 Cylinders
Order of Service
Cylinder
0 20 40 60 80 100 120 140 160 180 200